A classical segment for the local–global passage

From primewise neighbourhoods to a finite étale cover

31 October 2026

Part I

The mathematical argument

1 The precise classical principle

We work with commutative rings with identity, including the zero ring. In this part we assume classical logic and the following principle:

> Every proper ideal of a ring is contained in a prime ideal.

The principle is kept separate from classical logic. It can subsequently be deduced from choice by applying Zorn's lemma to the proper ideals containing a given proper ideal, and proving that a maximal ideal is prime. That derivation is not needed for the argument below.

Theorem 1. *Let E be a ring and let G be a property of its elements. Suppose that for every prime ideal $\mathfrak{q}$ there exists $s \in E$ such that $G(s)$ and $s \notin \mathfrak{q}$. Then there are finitely many elements $s_1, \ldots, s_m$, each satisfying G, and coefficients $a_i \in E$ with*

$$\sum_{i=1}^{m} s_i a_i = 1.$$

The family may be empty when E is the zero ring.

Proof. Let J be the ideal generated by all elements satisfying G. Membership in J means having a finite expression $\sum_i s_i a_i$ with $G(s_i)$. If $1 \notin J$, prime-ideal existence supplies $\mathfrak{q} \supseteq J$. The hypothesis gives a good s outside $\mathfrak{q}$, while $s \in J$ puts it inside $\mathfrak{q}$, a contradiction. Classical logic therefore gives $1 \in J$. Its finite expression is exactly the required equation. □

This proof never chooses a neighbourhood simultaneously for every prime. It considers just one prime containing J under the assumption that J is proper. The resulting equation is the algebraic certificate that the principal opens $D(s_i)$ cover $\operatorname{Spec} E$.

The property may mean that an element carries additional data. After the finite family is obtained, finitely many witnesses for that data can be retained along with the equation. This uses only finite choice, proved by induction on the family length.

2 Application to the local structure theorem

Fix a ring map $R \to E$. In the monic convention of *Local structure of étale algebras*, a standard étale algebra has the form

$$(R[T]/(P))[1/G(t)],$$

where P is monic and $P'(t)$ is invertible in the displayed algebra. Inverting $G(t)P'(t)$ instead gives the same kind of presentation.

Corollary 2. *Suppose that for every prime $\mathfrak{q}$ of E there exists $s \notin \mathfrak{q}$ such that $E[1/s]$ is standard étale over R. Then there are $s_1, \dots, s_m$ generating the unit ideal of E such that every $E[1/s_i]$ is standard étale over R. The standard presentations and localisation isomorphisms may be retained as part of the conclusion.*

Proof. Apply Theorem 1 to the property that $E[1/s]$ has such a presentation. Retain its finite family of presentation witnesses. □

There is a distinction between a neighbourhood standard over R and one standard only over $R_{\mathfrak{p}}$. To apply the corollary to the local theorem used in the ZMT argument, one first carries out the following algebraic steps.

Given $\mathfrak{q} \subset E$, put $\mathfrak{p} = \mathfrak{q} \cap R$. Classically $R_{\mathfrak{p}}$ is residually discrete local. The local theorem, after its hypotheses have been supplied, gives a principal neighbourhood standard over $R_{\mathfrak{p}}$. Clear the denominator of its defining element to obtain $s \in E \setminus \mathfrak{q}$.

Finite presentation is then used to descend the presentation and its inverse maps: all the data and identities involve finitely many coefficients. Consequently there is $f \in R \setminus \mathfrak{p}$ such that $E[1/sf]$ is standard étale over $R[1/f]$. The equations certifying the inverse maps and the inverse derivative must be included among the identities being descended.

Finally change the base back to R by monic rescaling. If the equation over $R[1/f]$ is $P(T)$ of degree $d > 0$, choose N large enough that

$$\widetilde{P}(Z) = f^{Nd} P(Z/f^N)$$

has a monic lift to $R[Z]$. If G is the polynomial being inverted, choose M such that $f^M G(Z/f^N)$ has a lift $V \in R[Z]$. Then

$$(R[1/f][T]/(P))[1/G(t)] \simeq (R[Z]/(\widetilde{P}))[1/(fV(z))].$$

The factor f in the denominator is retained. It makes the substitution $z = f^N t$ invertible and removes any ambiguity from the chosen lifts. The derivative becomes $f^{N(d-1)}P'(t)$, so remains invertible. The degree-zero case is the zero algebra and is treated separately.

This supplies the hypothesis of Corollary 2. Thus classical prime-ideal existence replaces the dynamical local–global argument; finite-presentation descent and the change-of-base lemma remain parts of the proof.

Part II
Connection with the formalisation

3 The classical segment and its parameters

The development is in `unit/sag/etale-classical-global`. The existing constructive imports remain in the usual `Theory` segment, with univalence, replacement and surjection parameters. Inside

it is the separate segment

```
Classical.Cl[lem,prime-existence].
```

Only imports within this segment can use these two parameters. Neither principle is added as a global axiom or to the constructive ZMT or local étale developments.

`lem` is excluded middle for propositions at each universe level. `prime-existence` asserts the mere existence of a prime above each proper proposition-valued ideal of each commutative set-ring. The type of this principle is defined in `PrimeExistence`, outside the classical segment. No inhabitant is asserted there.

A prime in this interface has propositional membership, properness and mere disjunctive primality. Decidability is not part of the existence principle. `ClassicalLogic.Prime` derives it from excluded middle and constructs the exact `PrimeIdempotents.Prime` record used by the earlier local proofs. Deriving prime-ideal existence from choice is left as a future theorem.

4 The checked compactness argument

The generic theorem is

```
ClassicalPrimeCover.Over[E,setE,G]
    .Given[pointwise].theorem.
```

Its premise is the truncated existence of a good element outside each prime. Its conclusion is the truncated existence of a natural number m, two families $s, a : \mathrm{Fin}(m) \to E$, witnesses of $G(s_i)$, and the equation $\sum_i s_i a_i = 1$.

The ideal of good elements is constructed using finite lists and propositional truncation. Its ideal laws and the inclusion of every good element are proved. A hypothetical proper ideal gives a prime above it; the primewise hypothesis contradicts containment. Double-negation elimination is applied only to the proposition $1 \in J$. The finite certificate is then extracted under truncation. Compactness itself is proved, rather than introduced as a further parameter.

Actual localisation certificates contain algebras and universal properties, so live one universe above the ring carrier. The variant `ClassicalCertifiedCover` handles this without a resizing axiom. For each s, excluded middle represents the proposition $\|\mathrm{Certificate}(s)\|$ by either the small type `Top` or the small type `Empty`, with maps in both directions. These small predicates generate the ideal. Once the finite family is found, ordinary finite choice restores its actual certificates inside the final truncation. This step uses no additional axiom of choice.

5 The étale endpoints and their scope

`ClassicalEtaleCover.Over[A,setA,E,setE,iota]` exports three adapters. Each takes a neighbourhood theorem for every decidable prime of E and gives a finite cover with its coefficient equation.

- `Given[local].theorem` accepts the exact conclusion of `Local.Statement`. Every chart includes an actual algebra T, the map $E \to T$, the universal property $T = E[1/s]$, and the existing `StandardEtale.Points.Standard` certificate.
- `GivenFlat[local].theorem` accepts the flat local conclusion. It additionally retains the square-zero lifting definition of étale for each chart.

- `GivenMonic[local].theorem` accepts and retains a stronger `MonicChart`: its polynomial is explicitly a coefficient list followed by the leading coefficient 1. This supplies the monic form of Corollary 2.

The last distinction matters because the older `Local.Statement` conclusion quantifies its polynomial Q but does not record monicity. The new ordinary adapter preserves that interface; it does not recover missing information. A forgetful map from `MonicChart` to the older chart interface is proved.

These are conditional local-to-global theorems: their neighbourhoods must already be standard over the original base A. This development does not construct the local setup at $A_{\mathfrak{p}}$, discharge the older local proof's remaining inputs, or formalise the finite-character and monic rescaling lemmas described in Part I. It also does not assemble a global square-zero lifting certificate from the finite cover. Accordingly it proves the classical compactness step, while the full global structure theorem from its original hypotheses still needs that integration. The constructive dynamical replacement remains separate.

6 Validation

A fresh check of `All.seg` gives 4262 declarations, no failures, in 1.49 seconds. The opening audit covers all 217 named declarations in `PrimeExistence` and `Classical`. Each opens at most 11 definitions and is reported at 0.00 seconds with two-decimal timing. The audit reports no openings outside the specified lists.

Regressions cover the good element 1, the empty cover of the zero ring, the impossibility of an empty-good cover in a nonzero ring, pairs $a + b = 1$ and complementary elements $e, 1 - e$, and certificates containing a universe-valued component. An actual standard algebra for $Q = X$, $G = 1$ is constructed over every base using `StandardAlgebra.Build`. Its identity localisation at 1 is proved by the universal property, and both the ordinary and monic cover theorems are applied to it.

Three mutations in isolated copies were rejected: changing the cover equation to a sum equal to zero, applying prime properness to the wrong element, and replacing the leading coefficient 1 by 0. The last is caught by the constructed-standard regression. Shared sources and the checker are unchanged; reused sources are relative links.

From `unit/`, check the development with

```
./segc sag/etale-classical-global all
./segc sag/etale-classical-global stop
```

The exact signatures and the fresh-check command are in `unit/sag/etale-classical-global/README.txt`.